MASTER'S THESIS DEFENSE · UNIVERSITY OF TRIESTE

Unsupervised State Bucketing for Mixture-of-Experts in Resource-Constrained Chess Engines

Can a tiny chess engine choose the right expert?

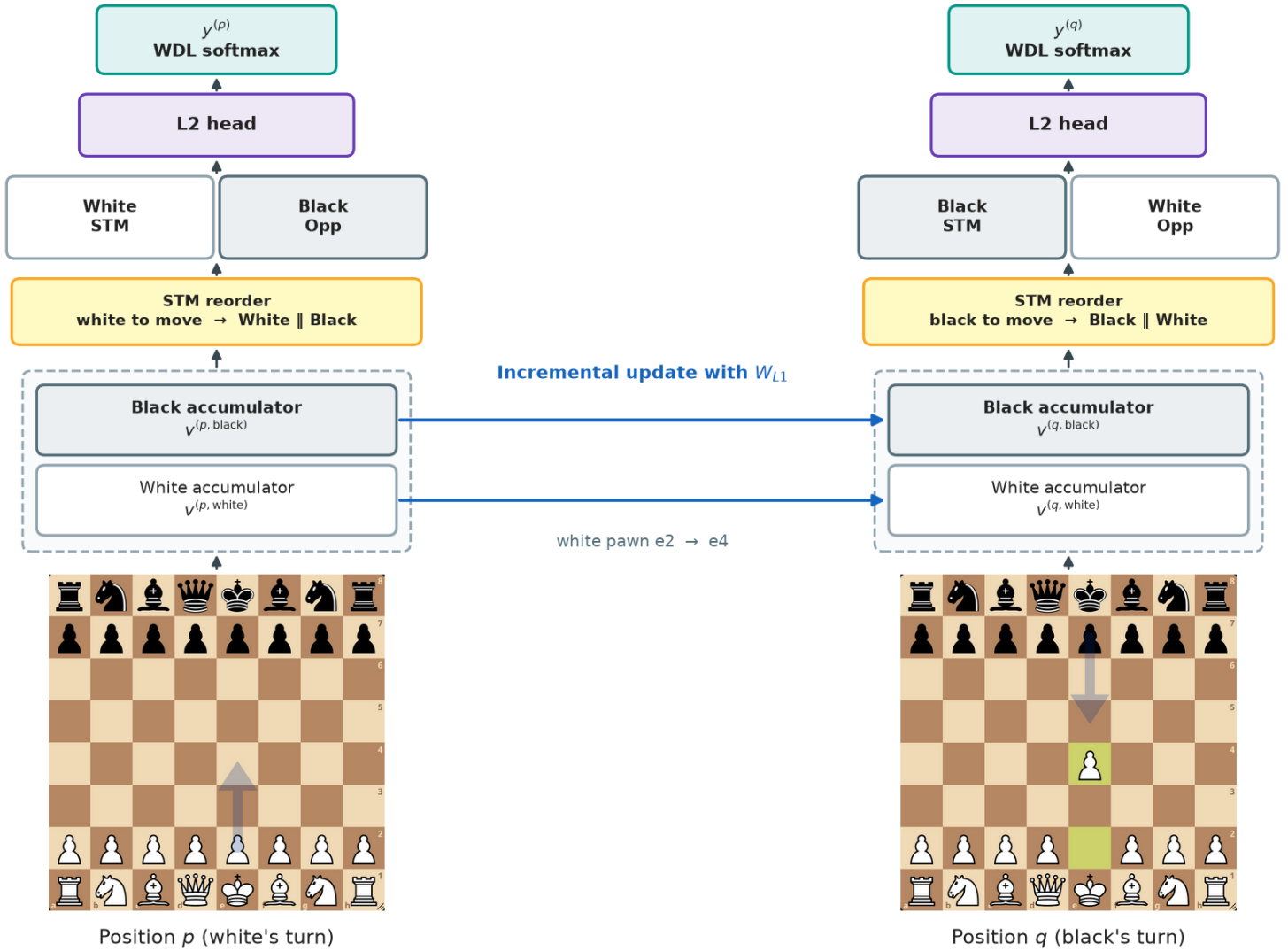
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Supervisor: Alessio Ansuini · Co-supervisor: Marco Zennaro, ICTP

The Big Picture: AI on the Edge

L1 is the shared first-layer representation throughout. Thesis Fig. 1.1.

The Architecture of the base NNUE



Position p (white's turn)

Position q (black's turn)

NNUE updates only the changed features

1

A move touches few inputs

Of the 844 board features, only the squares that change are active.

2

Edit the accumulator

Subtract the removed features and add the new ones. The rest stays.

3

Reuse it on the next node

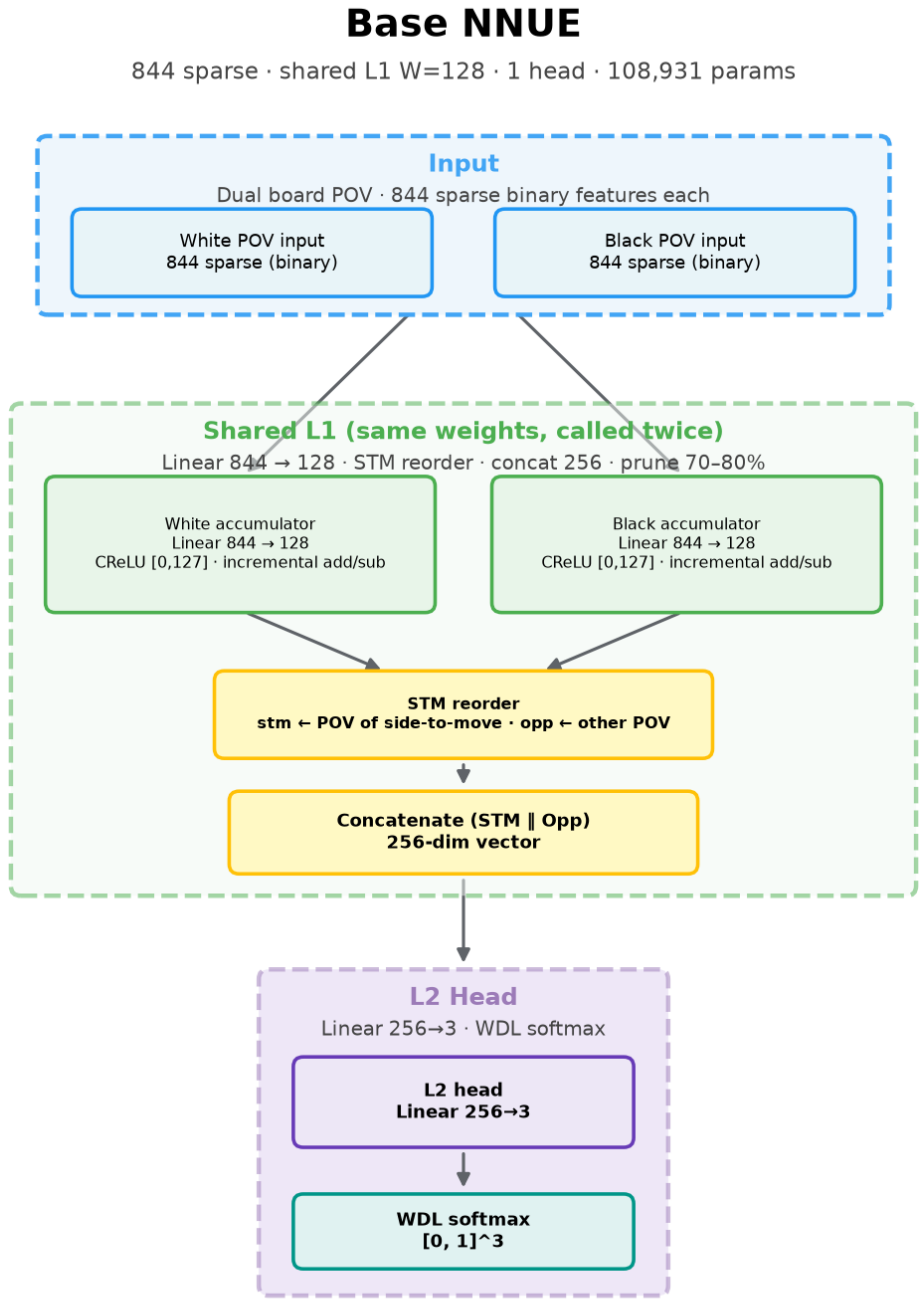
The L1 accumulator persists across moves, so evaluation stays cheap.

L1 is the shared first-layer representation throughout. Thesis Fig. 1.1.

NNUE ARCHITECTURE

The Architecture of the base NNUE

- Input layer
- Accumulator layer (L1)
- L2 hidden layer
- Output layer



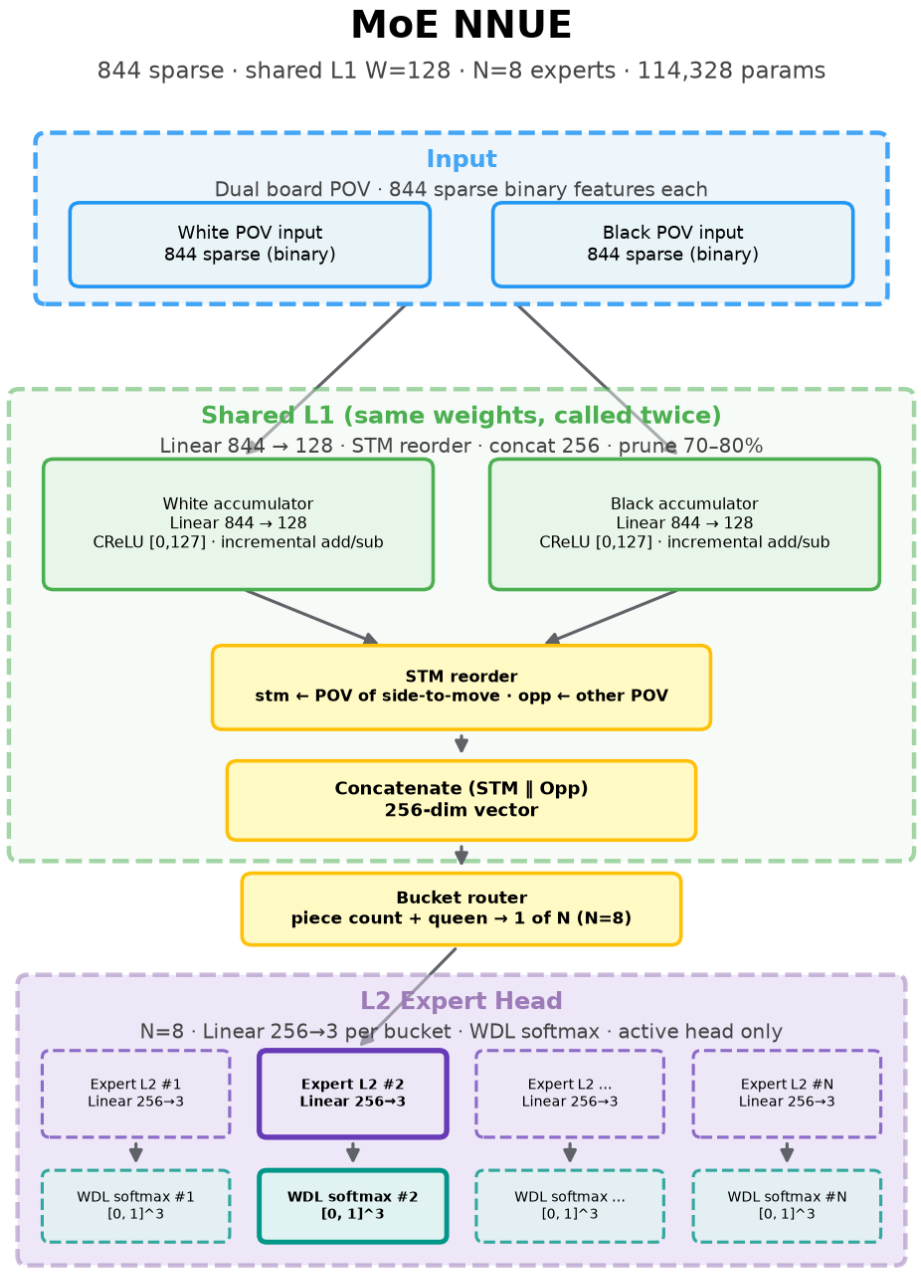
Student: DualHidden WDL · L1 W=64 · L2=128 · CE vs Lc0 STM (W,D,L)

encode_dual()

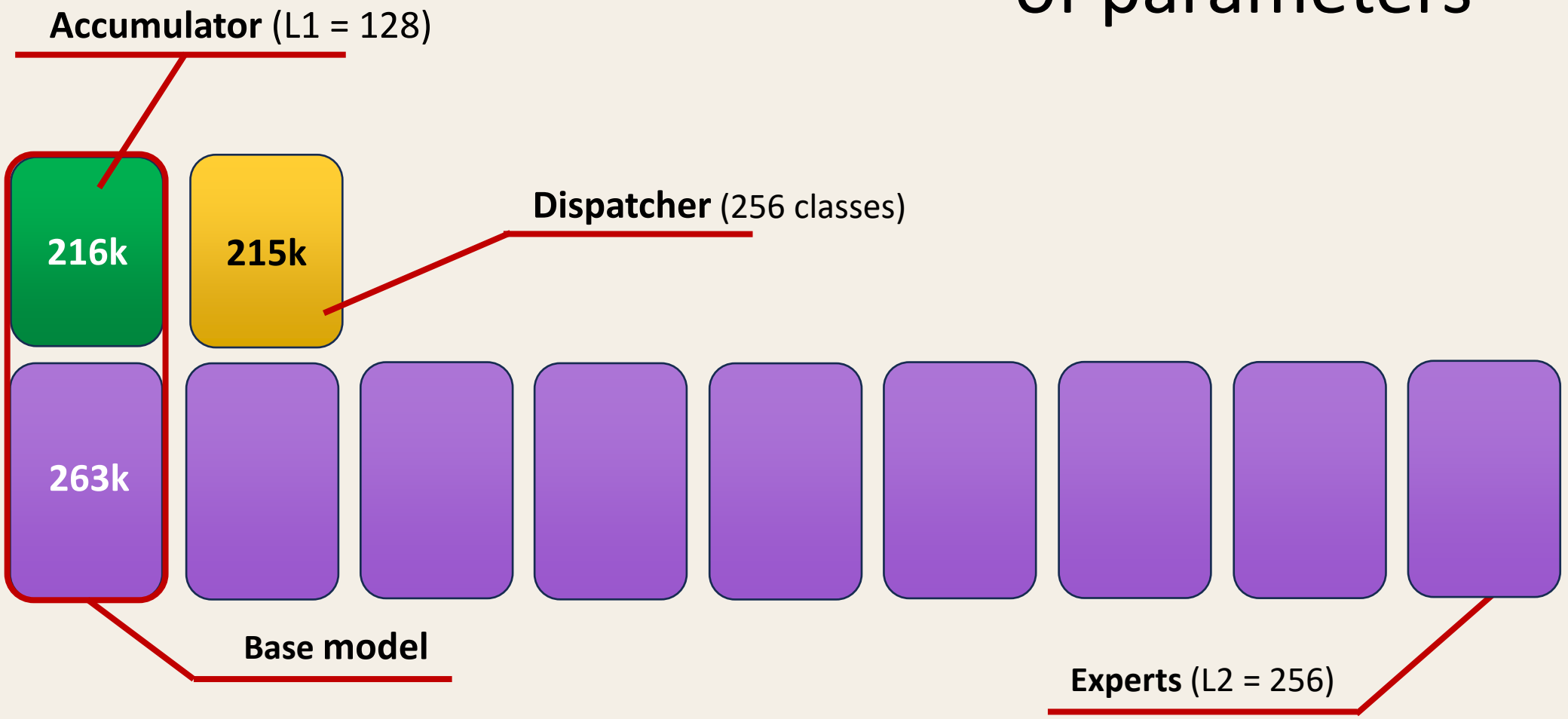
MoE NNUE ARCHITECTURE

The Architecture of the MoE NNUE

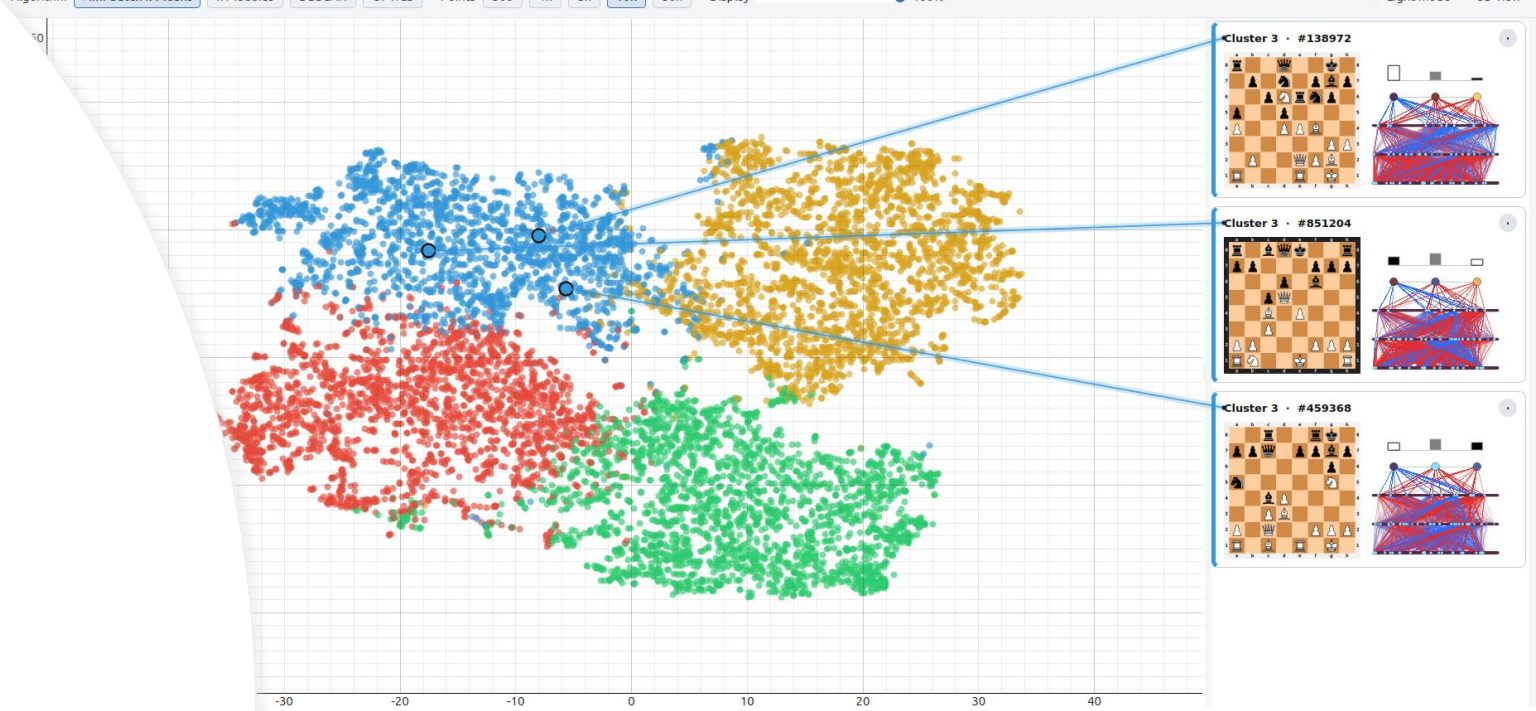
- A **router decides which head should be called**
- Expert heads can occupy a lot of memory
- During inference, only one head called (also the dispatcher)

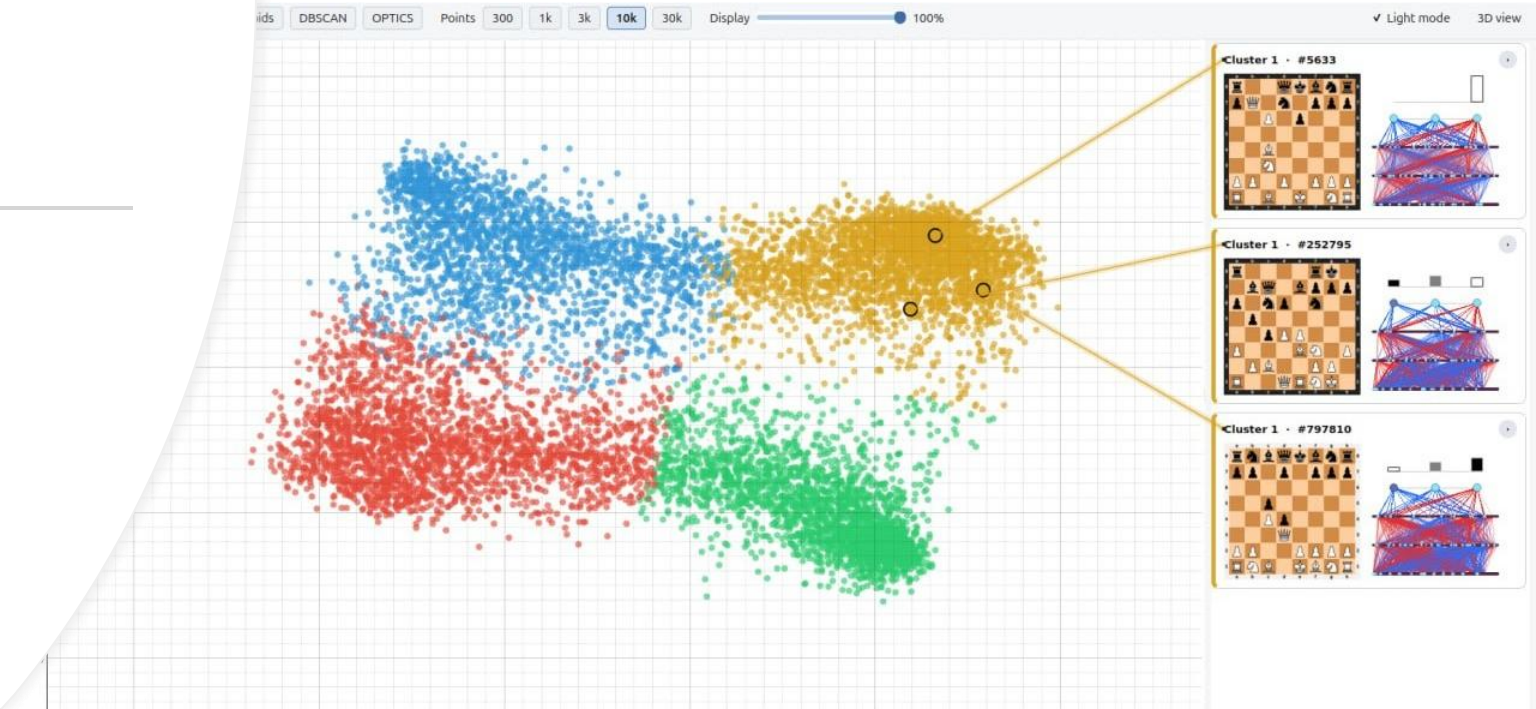


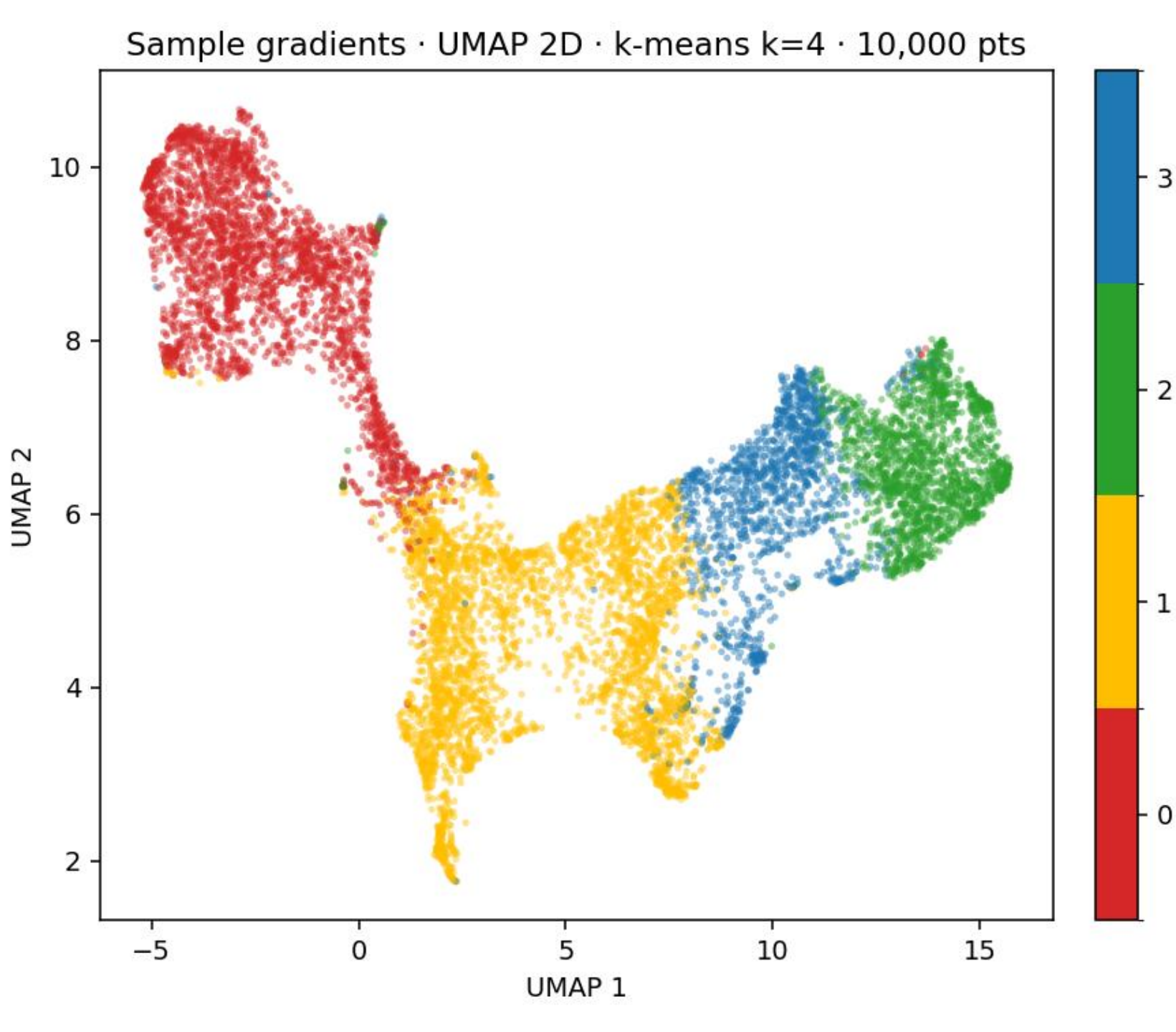
Number of parameters

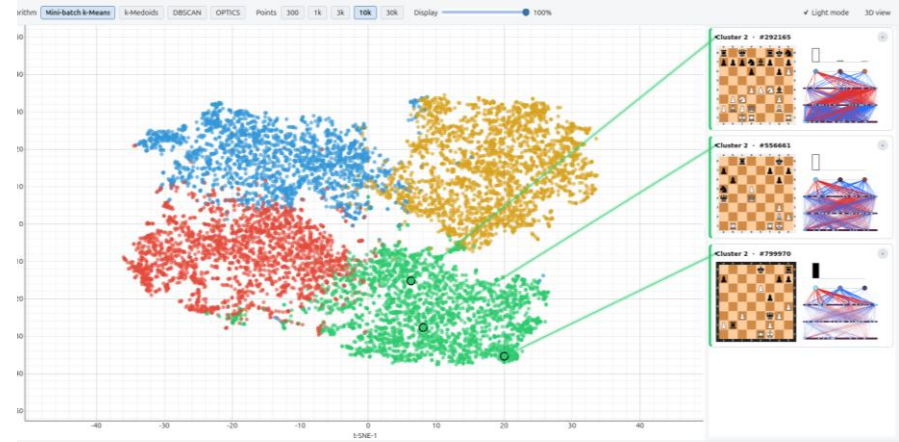


Clustering Sample Gradients






Sample gradients · UMAP 2D · k-means k=4 · 10,000 pts
UMAP 2
UMAP 1
10
8
6
4
2
−5
0
5
10
15
3
2
1
0







Training and evaluation protocol

150M

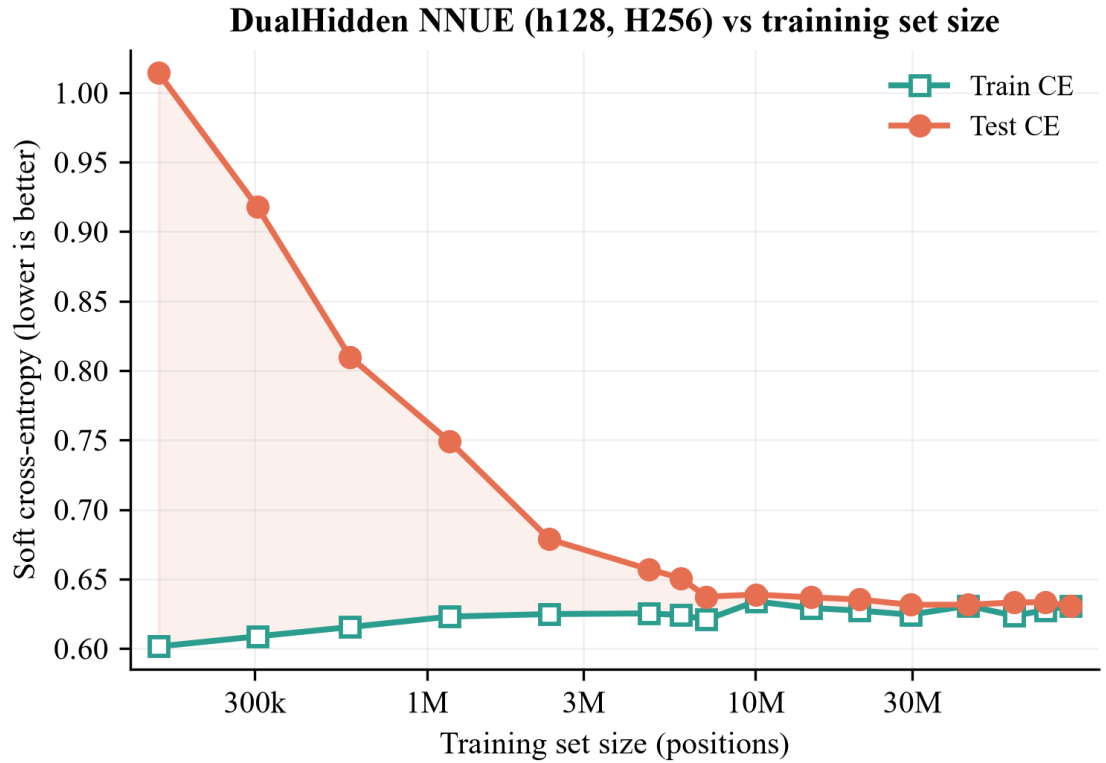
Training positions, mainly from Lichess.

~7M

Positions to overcome **overfitting**.

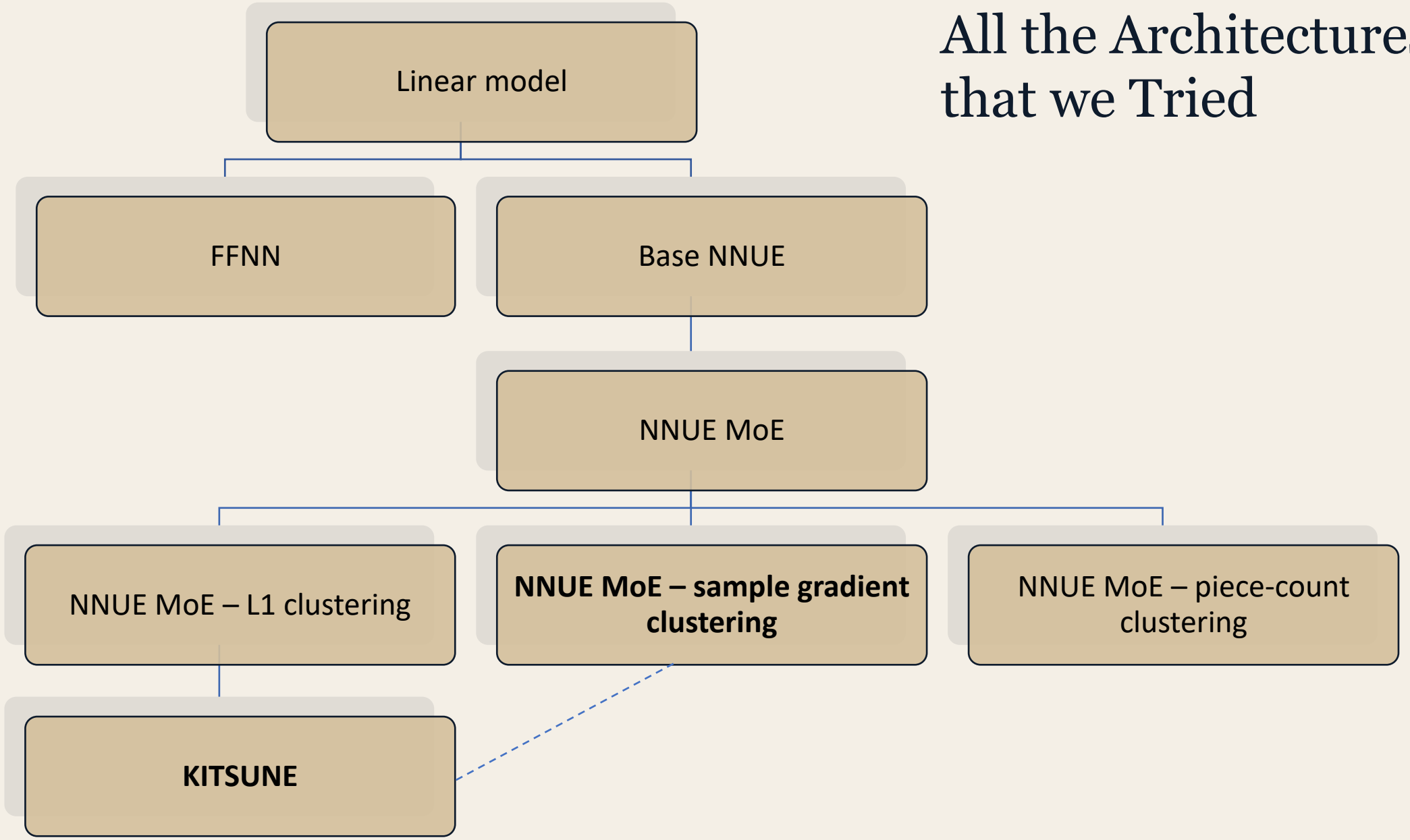
175k

Parameters in the biggest NNUE.



Lc0 supplies win / draw / loss targets. Lower cross-entropy is closer to the teacher. CE is not Elo. §§4.1–4.2, 5.1, 5.5.

All the Architectures that we Tried



SAME L1 INPUT, SAME MLP DISPATCHER, EIGHT BUCKETS

Gradient buckets are hard to recover



45.5 percentage-point gap

A training signal that depends on the true value is difficult to infer from L1 alone.

Tables 5.6–5.7. MLP width 64, K = 8, 1M-position cache, 90/10 split. Clustering was fit before the split.

Strong-base MoE gains are marginal

Model	K	CE
Reference NNUE	—	**0.6303**
Piece-count MoE	8	**0.6303**
L1-clustered hard MoE	16	**0.6300**
Gradient-clustered hard MoE	8	**0.6300**
Soft-gated MoE, top-2	2	**0.6304**

Best observed change in CE.

No decisive improvement over the reference NNUE.

Tables 5.10–5.11, 50,000 held-out positions. Lower CE is better. These figures are not an Elo gain.

SAME GRADIENT-TRAINED EXPERTS, DIFFERENT SELECTION RULES

The oracle exposes assignment headroom

K = 16

0.6301

Actual dispatcher

What can run online.

K = 16

0.6280

Gradient-centroid proxy

Still needs the teacher target.

K = 16

0.6005

Minimum-loss oracle

Each position goes to its best expert.

Table 5.13. The single-head base is CE 0.6303. Centroid proxy and oracle both need the unavailable teacher target.

Longer fine-tuning helps a weaker base

BASE · W64 / H128

0.6537

Test CE of the single evaluation head.

FOUR L1 EXPERTS

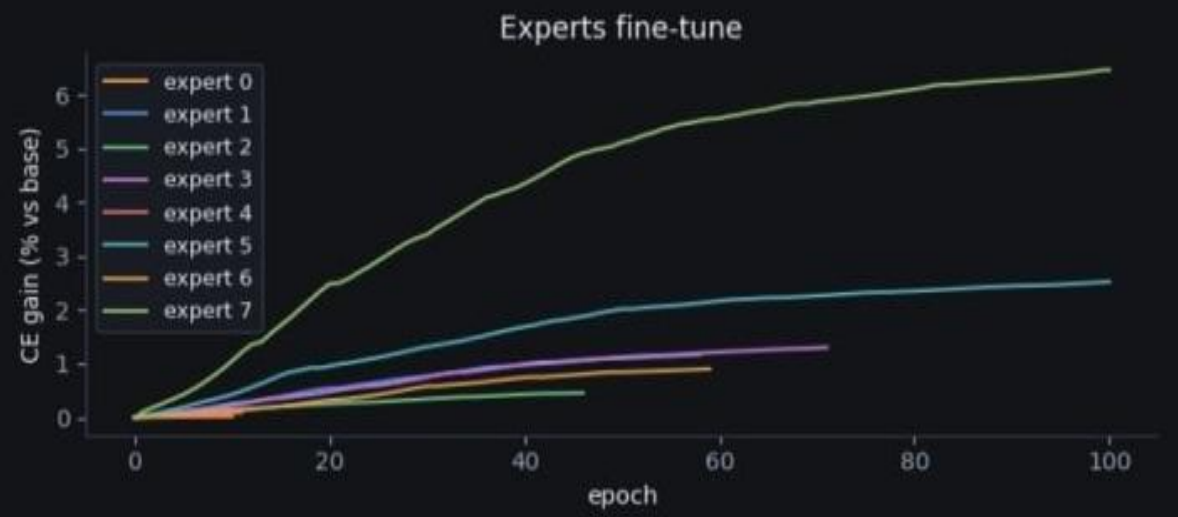
0.6499

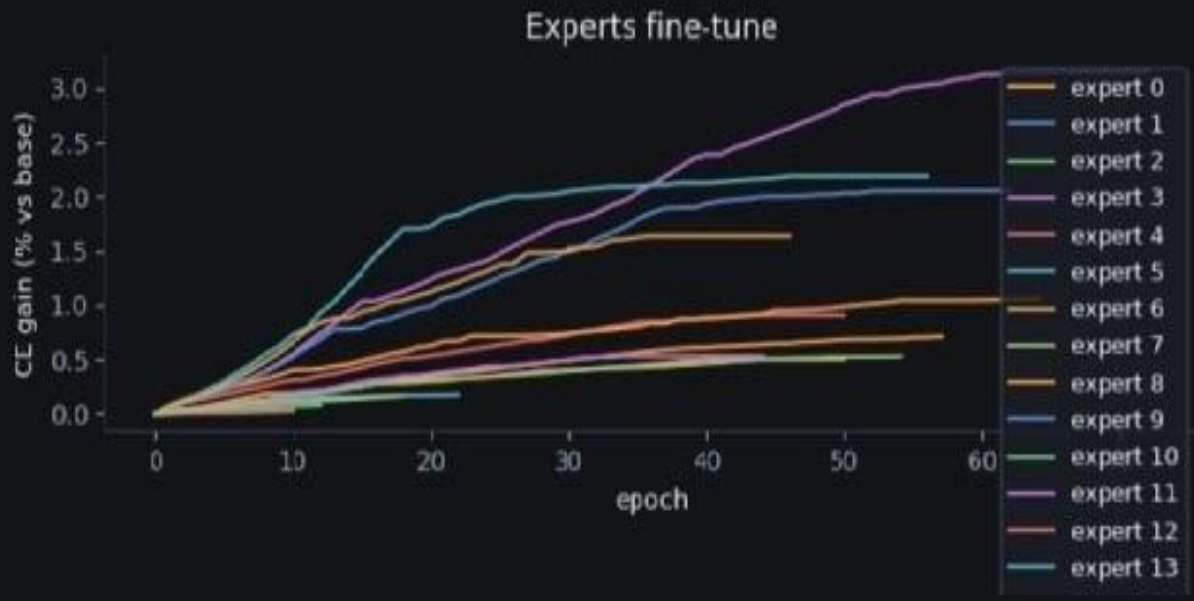
Linear dispatcher, longer fine-tuning.
A reduction of 0.0038.

This is the weaker base, not the strong reference. §5.5.4, K = 4. The gain depends on the model and the training budget.

Expert fine-tuning step

- Some partitions produce **very effective experts**
- Many partitions produce **modest improvements**
- The depth of the **loss basen** is rougly proportional to the initial magnitude of the **gradient**





KITSUNE stitches micro-buckets offline

1

Partition L1

Cut the activation space into many small, convex micro-buckets.

2

Stitch offline

Group those buckets by their mean head gradient. This step never runs during play.

3

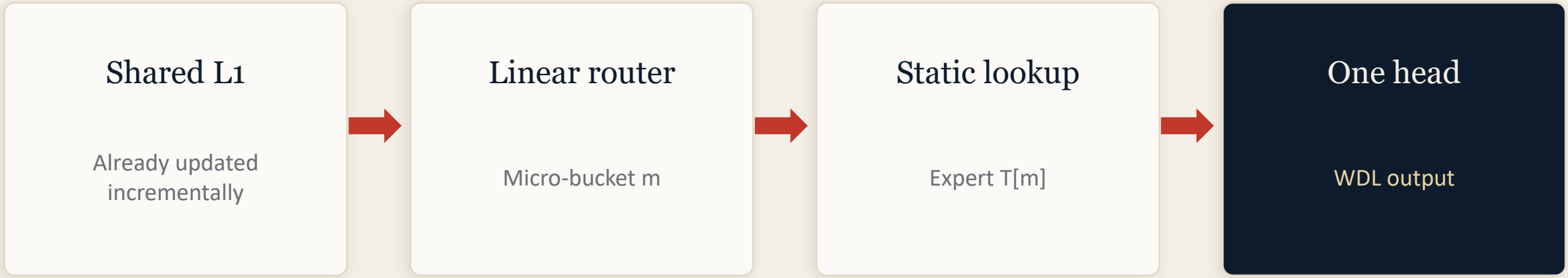
Train the experts

Disconnected L1 regions can share one expert, then each head is fine-tuned.

Knowledge-guided
Initialization and
Two-stage
Segregation for
Unsupervised
Neural
Experts

Thesis §3.3.5. Schematic, not a plot of measured geometry. Gradients shape the experts; they are not an online feature.

Online inference needs no gradients



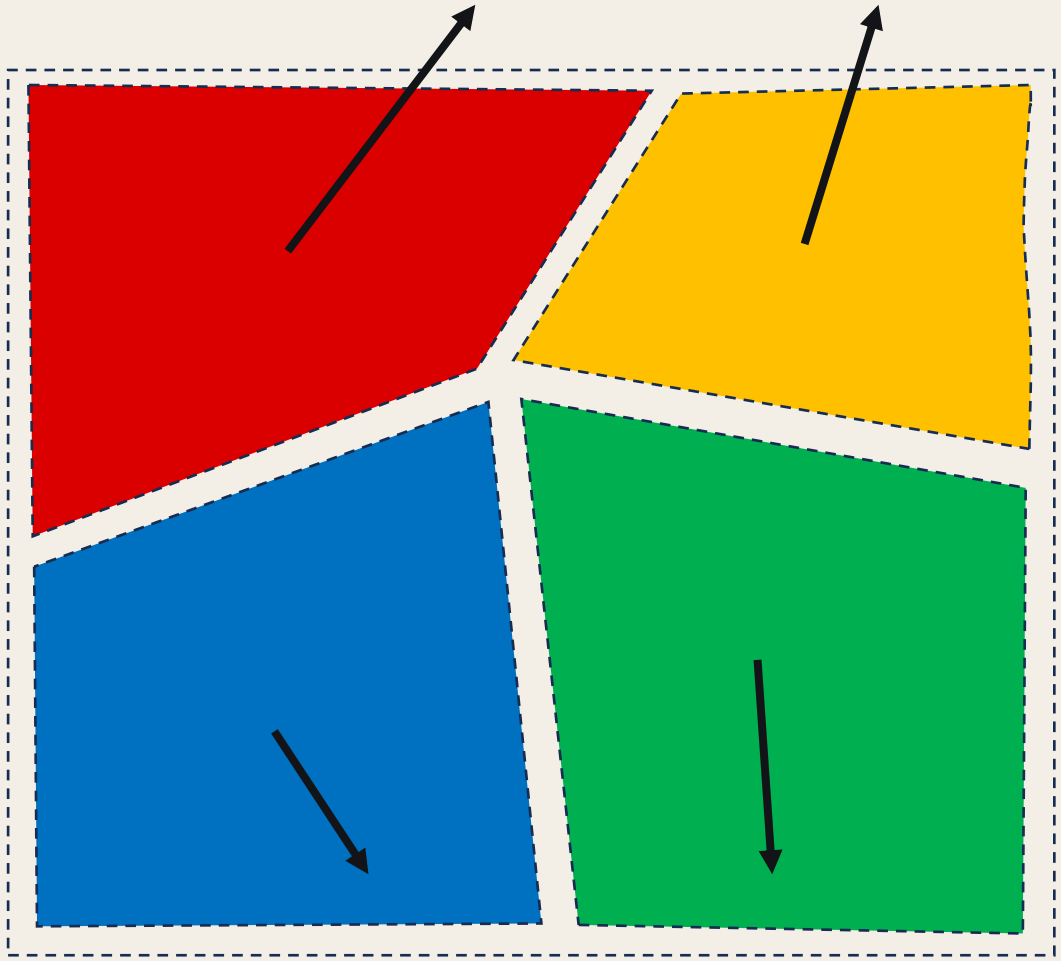
The online cost is a linear map plus a table lookup. Convex micro-buckets are what make that map enough.

The routed mixture on the weaker base is measured. It does not require a gradient at play time.

§3.3.5. Router cost scales with L1 width and the micro-bucket count. The lookup itself is a fixed table.

KITSUNE - Stitching

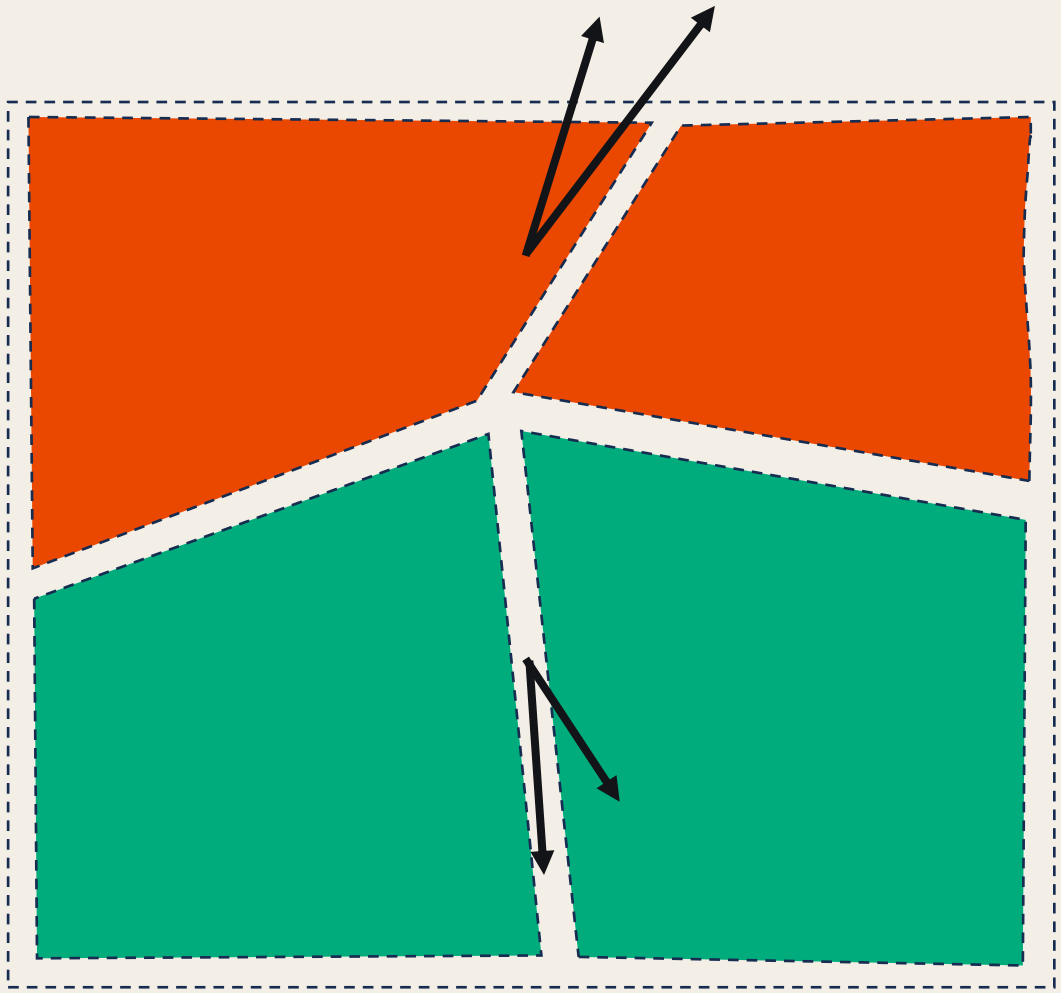
- You want B experts
- Partition the space to $B' = B\ k_{refinement}$ mini-clusters
- *Stitch* together the partitions with similar *average sample gradient*
- Accuracy vs B



Partitions of the L1 embedding space, and their respective average sample gradients.

KITSUNE - Stitching

- You want B experts
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Partitions of the L1 embedding space, and their respective average sample gradients.

KITSUNE matches that L1 result

0.65043

Routed mixture

K = 16, two micro-buckets per expert. 0.0033 below the base, and 0.0005 higher than the best L1 run.

0.60234

Best expert, own bucket

Factor 4. A 7.85% reduction against the base CE. Factor 2 is already at 0.60571.

Refinement factor at 16x

Still no significant improvement

At K = 16 the mixture rises to 0.65138 and the best expert to 0.60865.

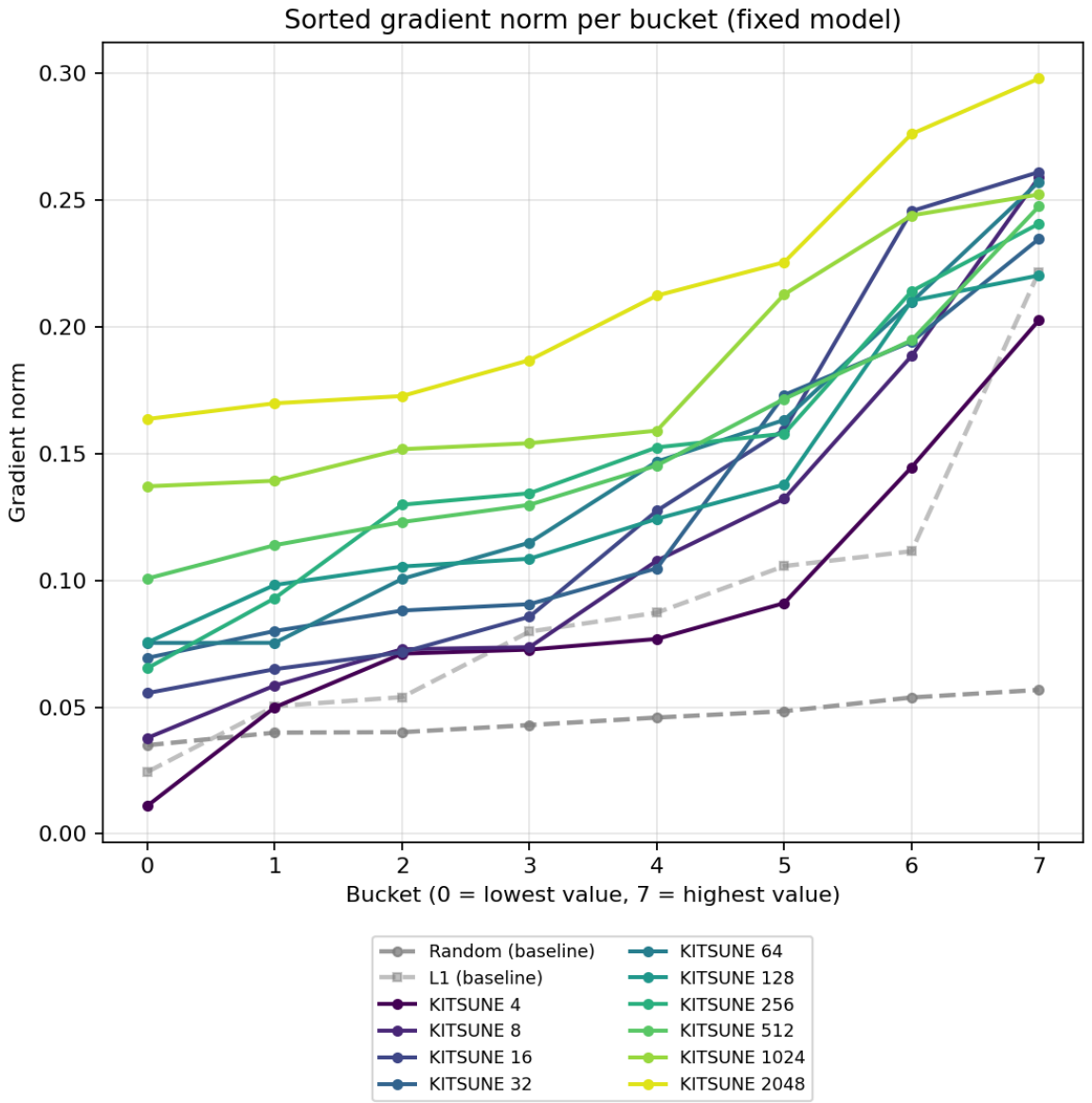
K	B′/K	MoE CE	Best expert	Gain
4	4	**0.65118**	**0.63726**	2.51%
4	8	**0.65137**	**0.64050**	2.01%
8	4	**0.65102**	**0.61748**	5.54%
8	8	**0.65093**	**0.61275**	6.26%
16	2	**0.65043**	**0.60571**	7.34%
16	4	**0.65061**	**0.60234**	7.85%
16	8	**0.65138**	**0.60865**	6.89%

Table 5.16. Expert gain uses the full-base CE, not a same-bucket comparison. Strong-base KITSUNE, Elo, and nodes per second are still open.

Some Last-minute Results...

$k_{refinement}$ needs to be **way bigger**!

- Less points in the cluster = more aligned sample gradients
- Downside: the dispatcher gets really big! … but that is OK.
- *Stitch* together the partitions with similar *average sample gradient*
- Accuracy on B' is not important



MAIN CONCLUSION

Useful experts need a cheap, reliable assignment.

Online

KITSUNE routes with L1 and a lookup. Gradients are used only while the buckets are stitched.

Mixture

On the weaker base, routed CE 0.65043 matches the best single-stage L1 result, 0.6499.

Experts

The best head, on its own bucket, reaches 0.60234. Past four micro-buckets per expert, both numbers worsen.

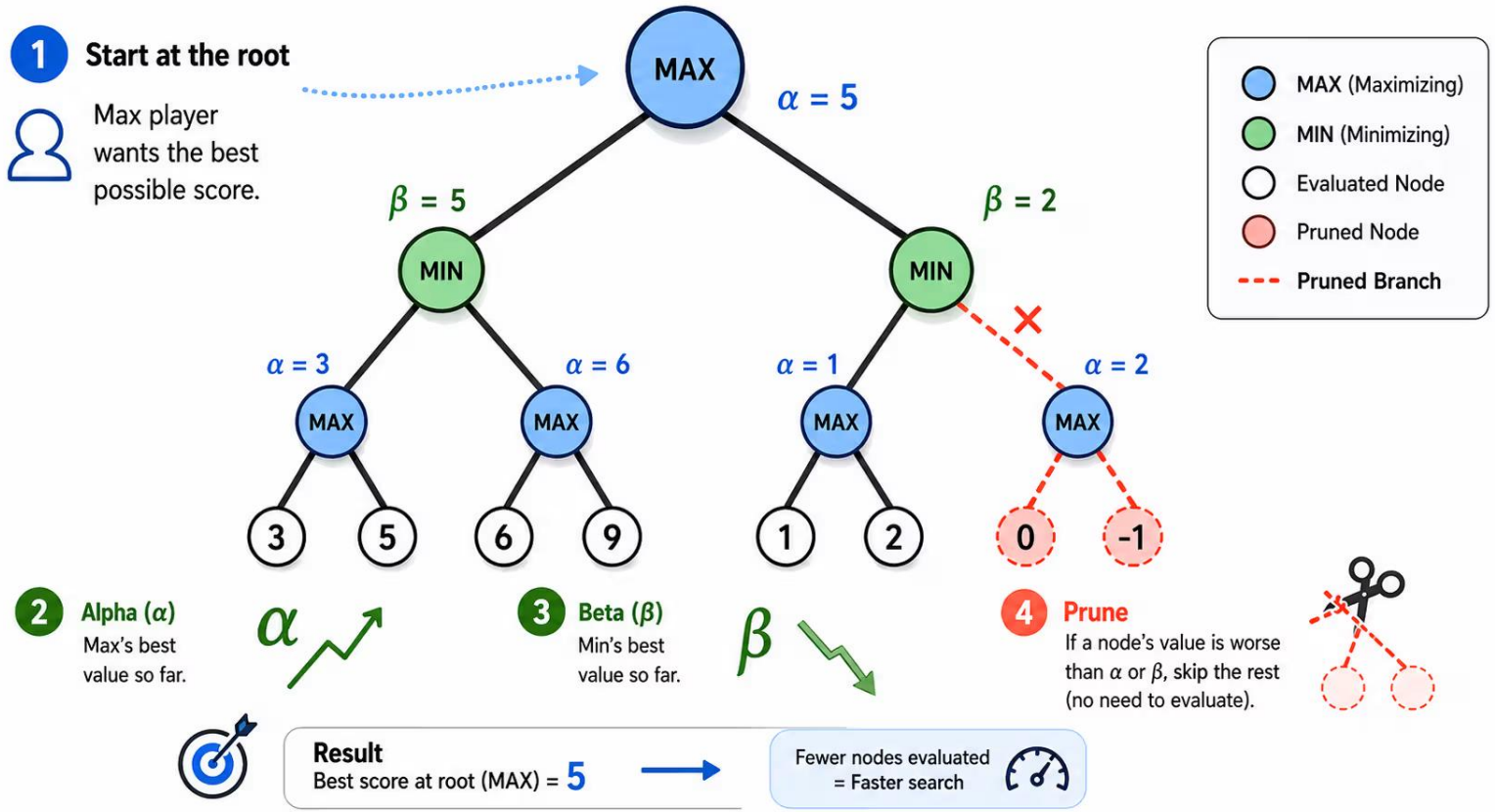
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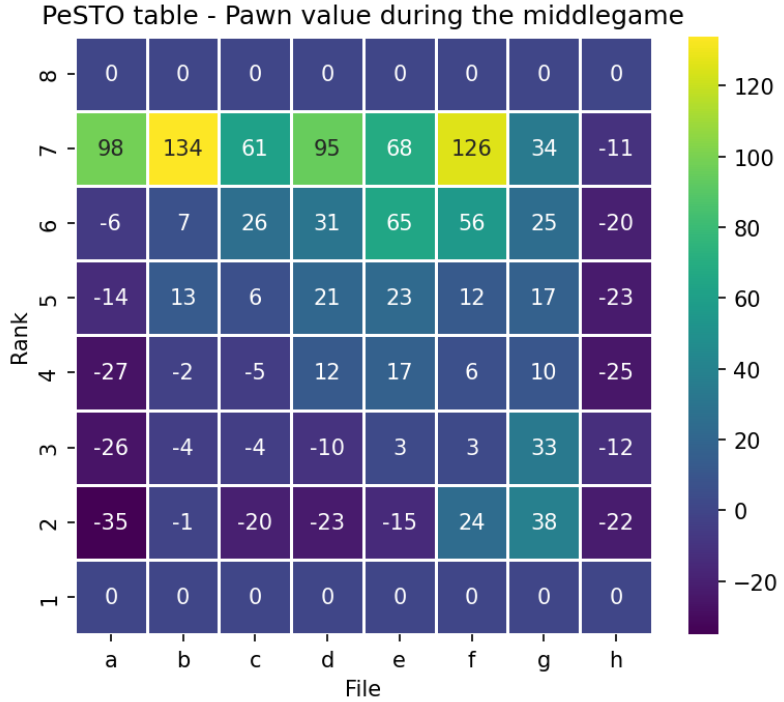
Thank you.

Questions?

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1 Start at the root
Max player wants the best possible score.
MAX
α = 5
β = 5
β = 2
MIN
MIN
α = 3
α = 6
α = 1
α = 2
MAX
MAX
MAX
MAX
3
5
6
9
1
2
0
-1
MAX (Maximizing)
MIN (Minimizing)
Evaluated Node
Pruned Node
Pruned Branch
2 Alpha (α)
Max's best value so far.
3 Beta (β)
Min's best value so far.
4 Prune
If a node's value is worse than α or β, skip the rest (no need to evaluate).
Result
Best score at root (MAX) = 5
Fewer nodes evaluated = Faster search


PeSTO table - Pawn value during the middlegame
Rank
File
8 0 0 0 0 0 0 0 0
7 98 134 61 95 68 126 34 -11
6 -6 7 26 31 65 56 25 -20
5 -14 13 6 21 23 12 17 -23
4 -27 -2 -5 12 17 6 10 -25
3 -26 -4 -4 -10 3 3 33 -12
2 -35 -1 -20 -23 -15 24 38 -22
1 0 0 0 0 0 0 0 0
a b c d e f g h
120
100
80
60
40
20
0
−20

THE DEPLOYMENT CONSTRAINT

Specialization must preserve search speed

120 MHz	192 KB	512 KB
CPU frequency	Internal RAM	Internal flash

Alpha–beta search calls the evaluator over and over. A slower network can cost a ply of search, which often matters more than a small gain in the value itself.

Target: Wio Terminal, ARM Cortex-M4.

The question

Can specialization improve the prediction inside this budget?

Hardware specifications, not measured deployment. Thesis §4.8.

OFFLINE STRUCTURE

Gradients form more separated clusters

Silhouette at K = 8, each score measured in its own representation.

0.191

Sample gradients

The most separated of the three signals.

0.003

L1 activations

Almost no silhouette. Easy to draw, little separation.

Separation is not the same as a better expert or a usable router. Table 5.2; gradient UMAP in Fig. 5.3.

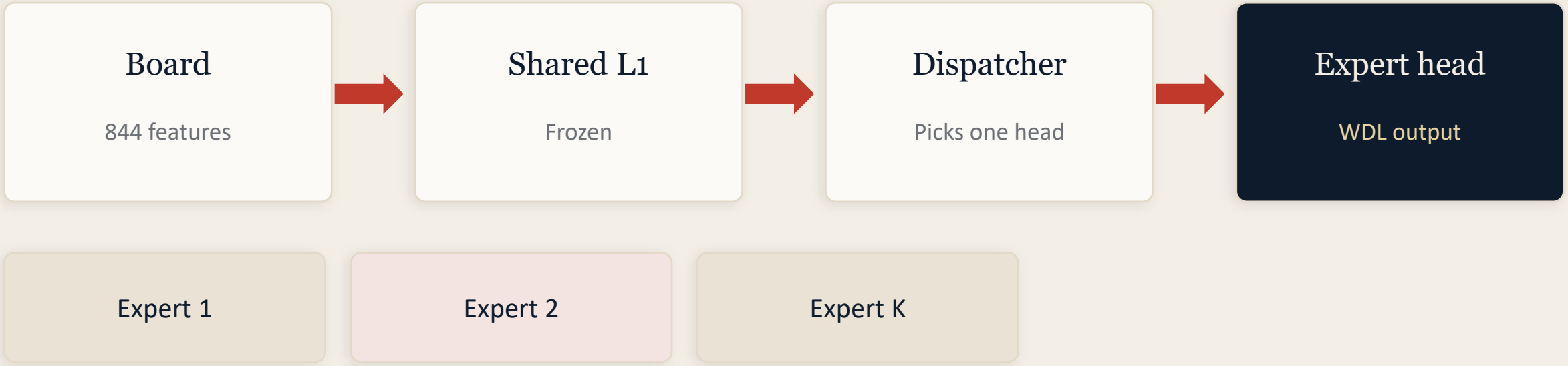
Bucketing defines each expert's task

Bucket signal	Groups positions by	Available online?
Piece count	A hand-designed game-phase proxy	Yes — a fixed rule
L1 activations	Similar hidden representations	Yes — already computed
Sample gradients	Similar updates of the head	No — needs the teacher WDL
Jointly learned gate	An end-to-end routing objective	Yes — after training

L1 describes the position. Gradients describe how the head should change.

Thesis §3.3. Per-sample head gradient uses the teacher target, which is unknown at play time.

One accumulator, one selected expert



Fine-tuning stores several small heads. At inference, L1 activations select one of them. The others stay in memory and do not run.

Hard routing, thesis §§3.3–3.4. Extra experts still cost memory.